

G05 / ASSESSMENT

G05 GATE B

Standalone Zero-to-Job-Ready Data Engineer 2026

Study, practise, check and explain.

A complete learning route

Start with the lesson video. Read the worked explanation beside it, pause to reproduce the example, then solve the supplied task. The books contain your study notes; keep your own work and results as evidence.

Use the protected answers after saving your first attempt. A changed retry shows what you can now do independently. There is no forced timer during learning.

One course. M00-M19. 264 lessons and 50 project phases.

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Read this first

Open only if routed after Review A repair. This is a different scenario, not a copy of Gate A. Save the attempt before opening Review B.

Fresh Case B - marketplace price stream

1. Architecture: Price-change events arrive through Kafka and feed a Delta lakehouse plus real-time consumer. Explain the responsibilities of each layer.
2. Kafka design: Choose a key that preserves per-product change order while avoiding unnecessary seller-wide ordering.
3. Consumer groups: Analytics and notification services must each see every event; notification has 6 replicas. Design group IDs and partition count reasoning.
4. Delivery: One price event is replayed after a crash. Design the sink so the business price changes once.
5. Schema: Evolve currency from implicit INR to an explicit optional field without breaking old consumers; then describe a deliberately breaking v3 change.
6. Spark: Explain how a new Structured Streaming query with no checkpoint uses startingOffsets differently from a restarted query with a checkpoint.
7. Operations: All partitions show rising lag after a deployment and rebalances are frequent. Give a diagnostic/recovery order.

Practical evidence B

1. Use G05_B_events.jsonl. Identify ordering-sensitive keys, duplicate IDs and one schema-evolution issue.
2. Design a replay into an isolated corrected Delta target and state the reconciliation before promotion.
3. Write the teach-back: explain in plain language why "exactly once" must always name its boundary.

G05 FRESH GATE B - REQUIRED IMPLEMENTATION

Required runnable integration evidence B

Implement the fresh marketplace case from G05_B_events.jsonl in a separate target/output. Do not reuse Gate A output/state.

Required behavior: product key preserves per-product ordering reasoning; duplicate P2 is idempotent at the business sink; P3 with price="130 INR" is rejected/quarantined under an explicit schema/validation rule; valid v1 records without currency follow your documented compatibility/default policy.

Create G05_B_RUNTIME_EVIDENCE.json with input count, valid/quarantine counts, unique applied event IDs, final price per product, schema-version/compatibility note, and replay result.

Deliberate failure: fail the target write after reading a valid event. Prove the event remains recoverable and is not committed/quarantined as poison data. Recover and rerun.

Changed retry: add P4 plus a newer SKU1 price and rerun with review closed. Reconcile before promotion.

Critical failure rules B

Fail if a valid failed sink write is committed, a duplicate changes price twice, the malformed price is silently coerced, or replay cannot be reconciled.

Keep the fresh A/B separation: Gate B is not a memorized copy of Gate A.